Utrecht University

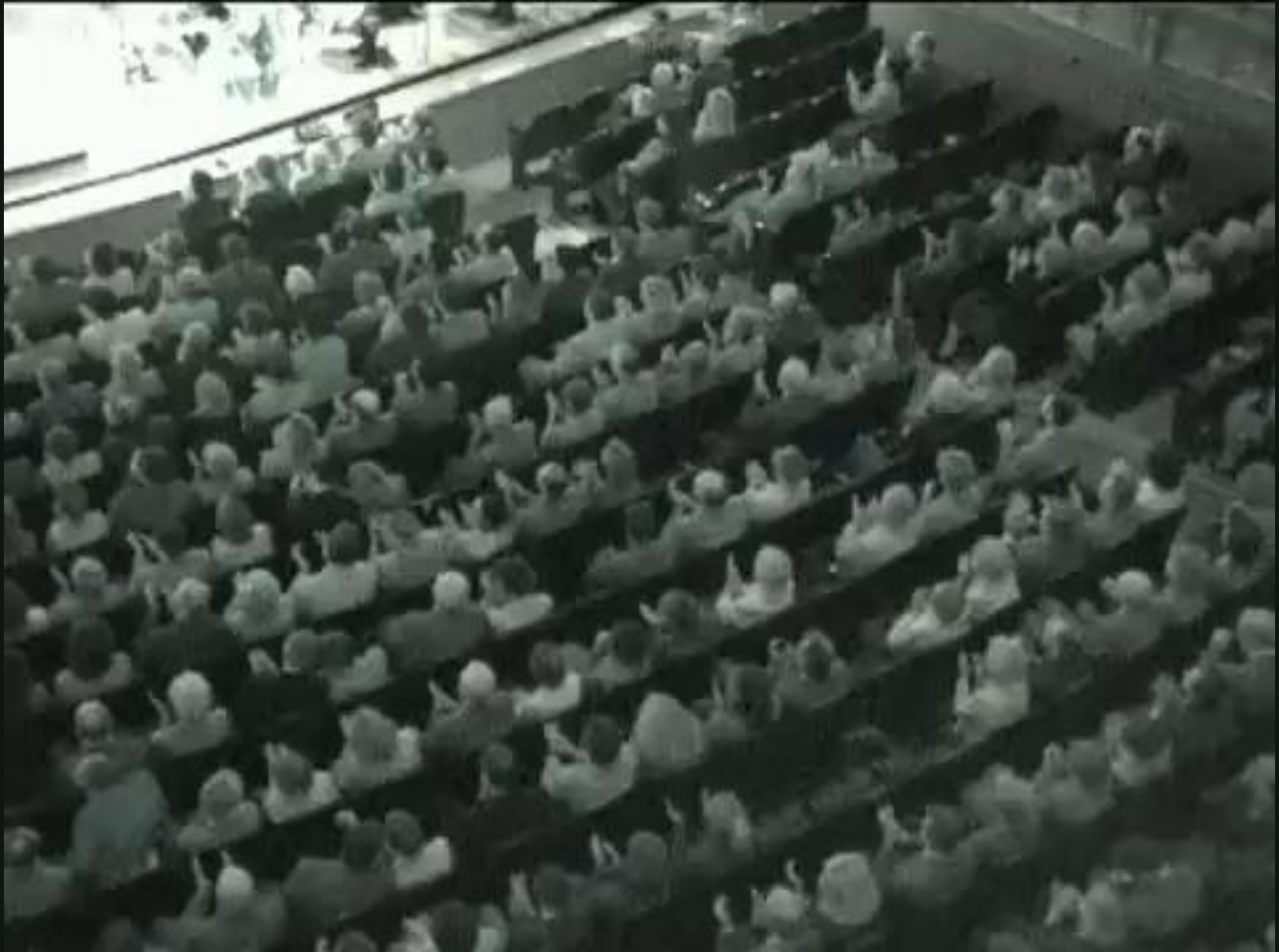
SUMMER SCHOOL - INTRODUCTION TO COMPLEX SYSTEMS

Feedbacks & Emergence

Dr. Robbin Bastiaansen
r.bastiaansen@uu.nl

Slides available at:
bastiaansen.github.io/talks/CCSS_SS_2026_Emergence.pdf

Part I Examples & Definition



Some definitions

A **complex system** is a (typically large) collection of (typically similar) elements that are connected to each other by ‘signals’.

‘signals’ could be e.g. money, energy, agreements, hormones, information, tweets, infection, friendship

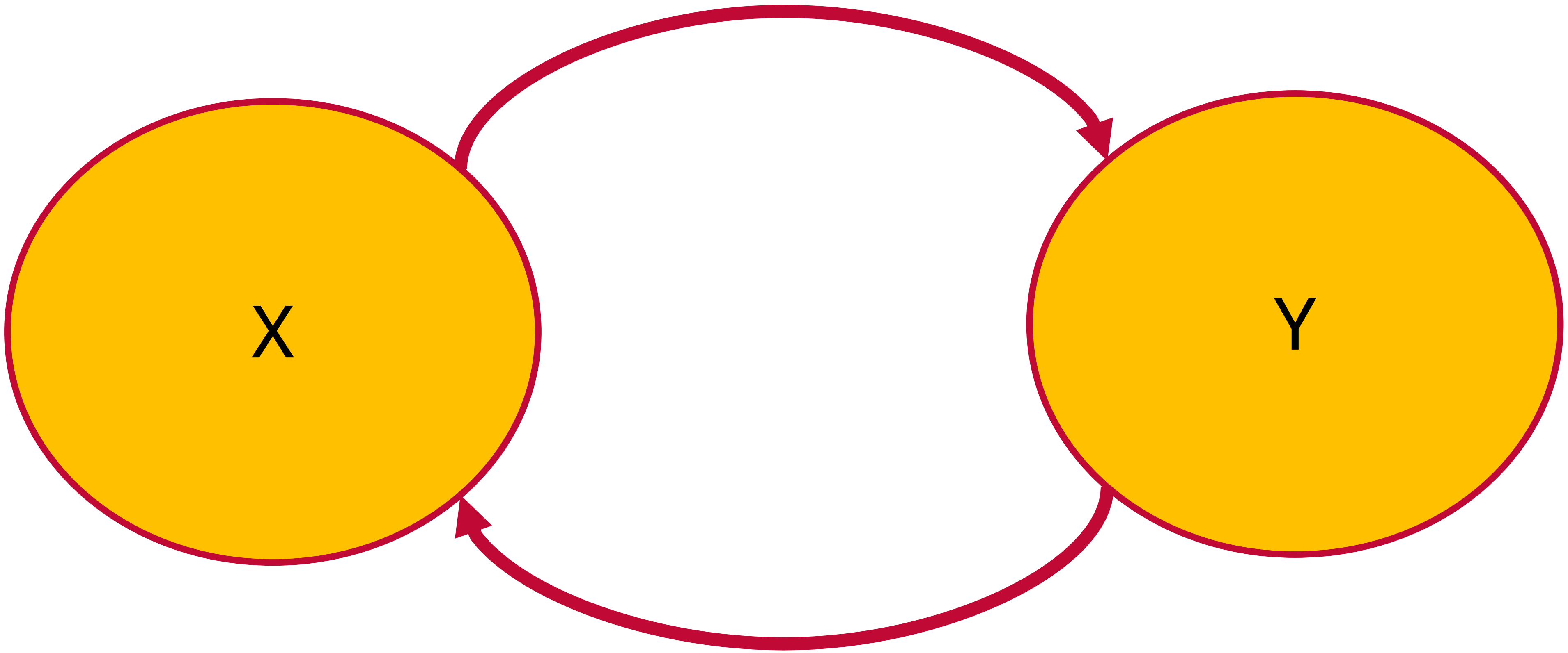
Emergence

Indivuals perceive signals from other individuals. Then, a sufficiently summed strength of these signals causes individuals to act in certain ways that produce additional signal, leading to emergent collective behaviour.

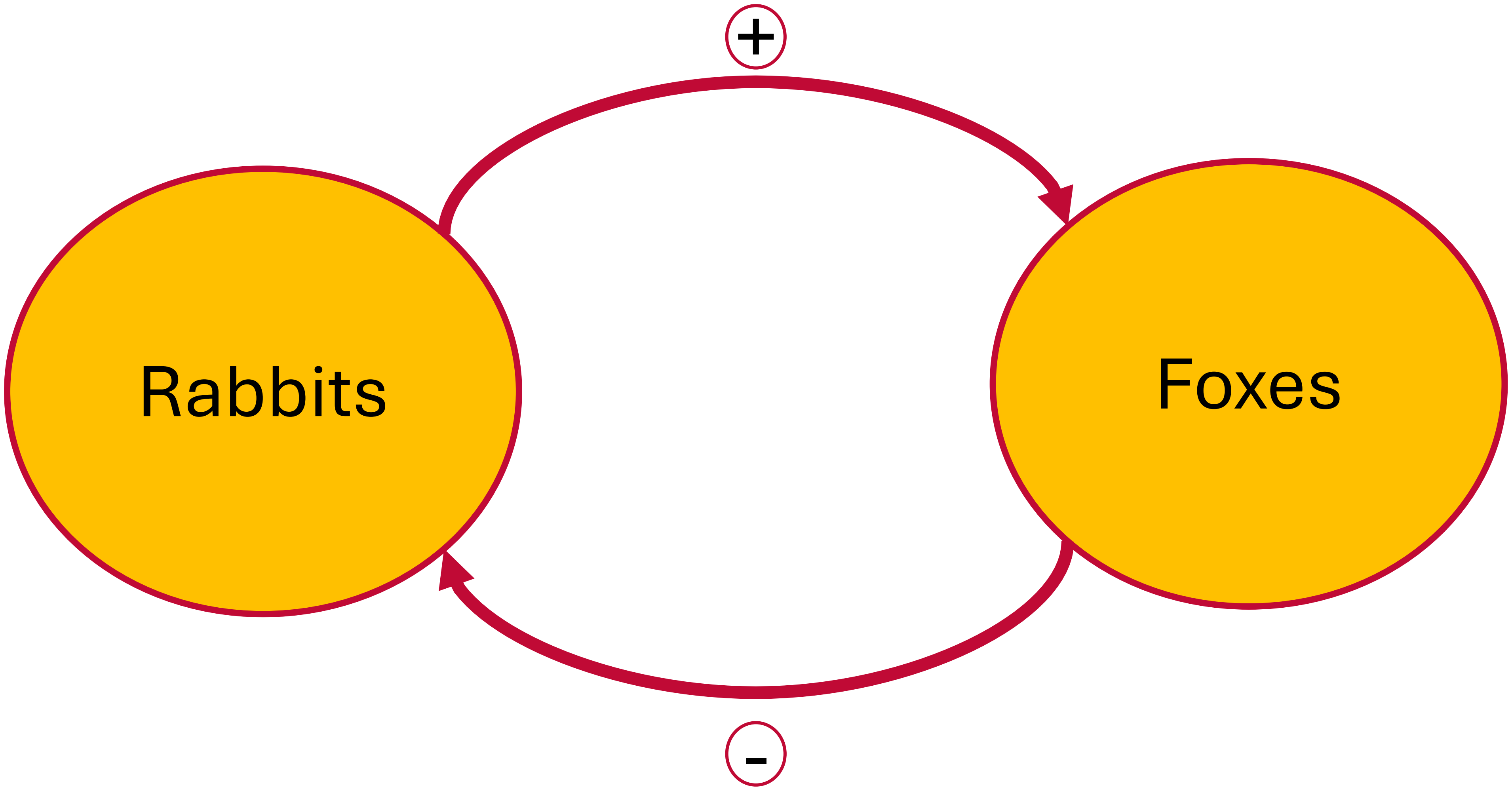
Crucially, NO central control. It is self-organised!

Part II
Feedbacks

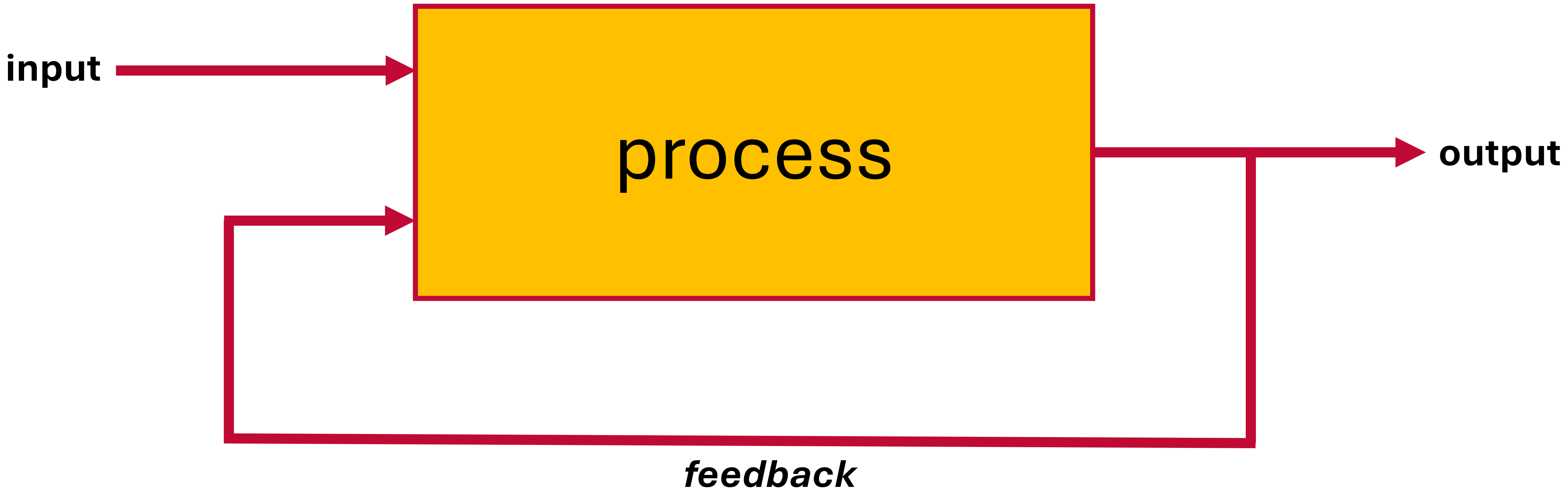
Feedback loops



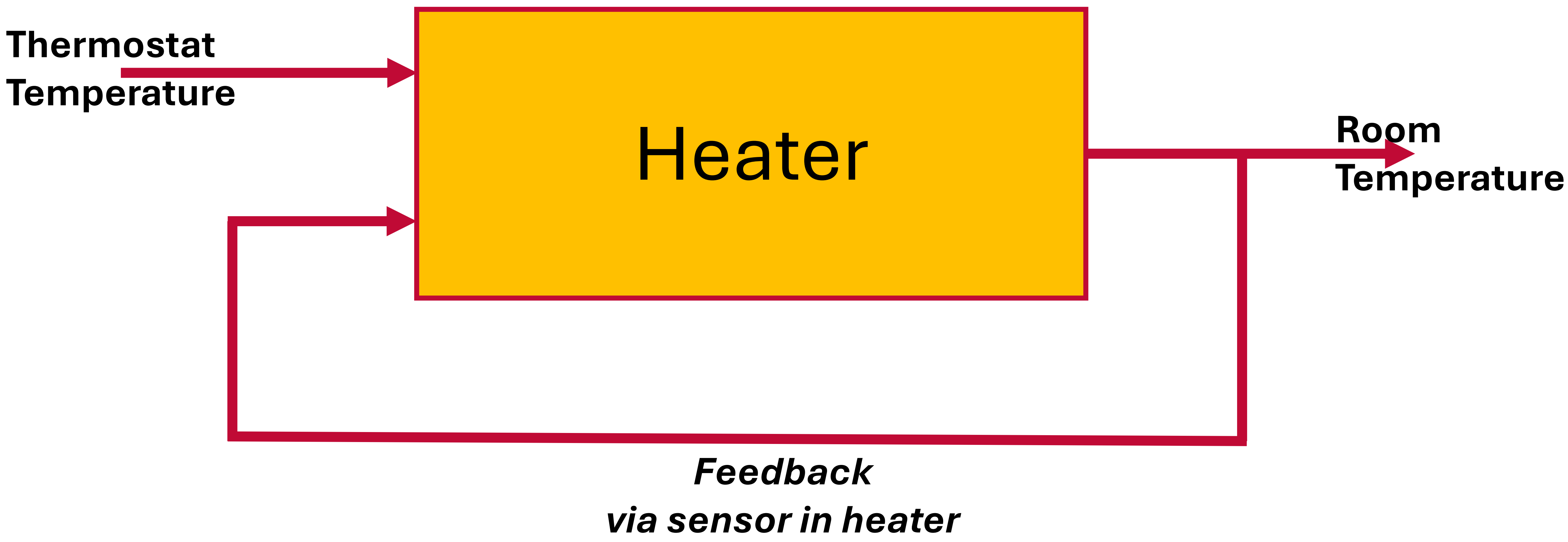
Feedback loops – Example predator-prey system



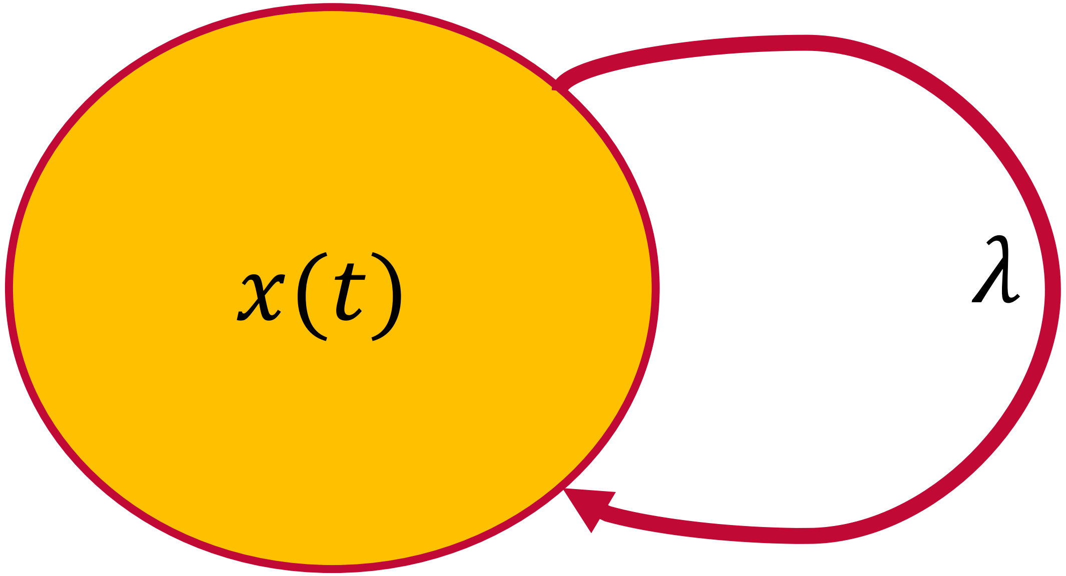
Feedback loops – classic control theory view



Feedback loops – classic – example



Dynamics



Generalises with multiple components:

$$\frac{d\vec{x}}{dt} = A\,\vec{x}$$

and eigenvalues of matrix A

$$\frac{dx}{dt}(t) = \lambda\, x(t)$$

$$x(t) = e^{\lambda\, t}\, x_0$$

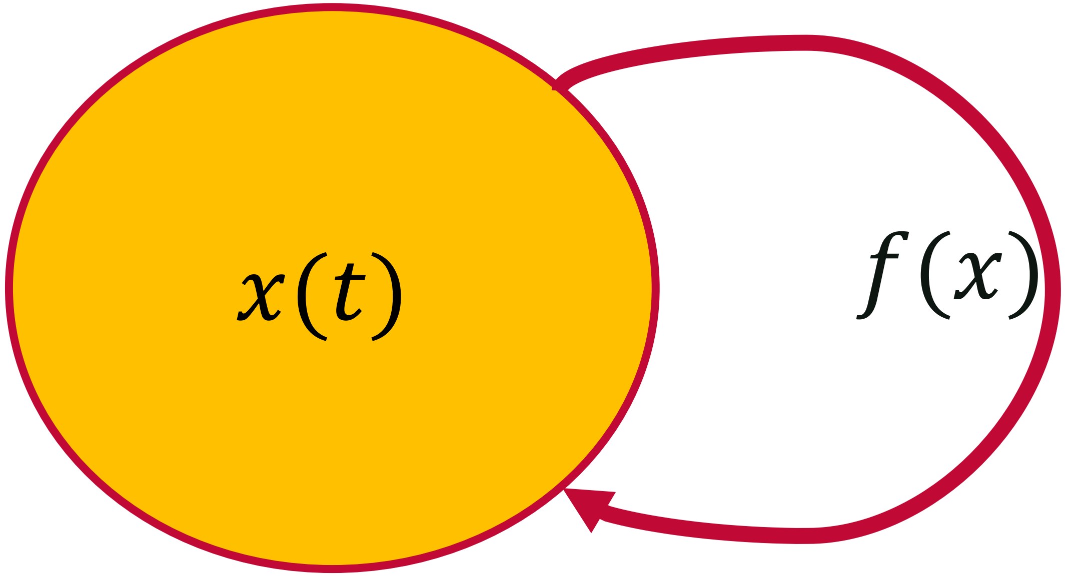
$$Re\,\lambda < 0 : x(t) \to 0$$

negative feedback

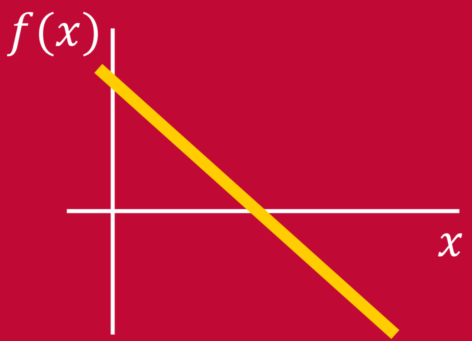
$$Re\,\lambda > 0 : x(t) \to \infty$$

positive feedback

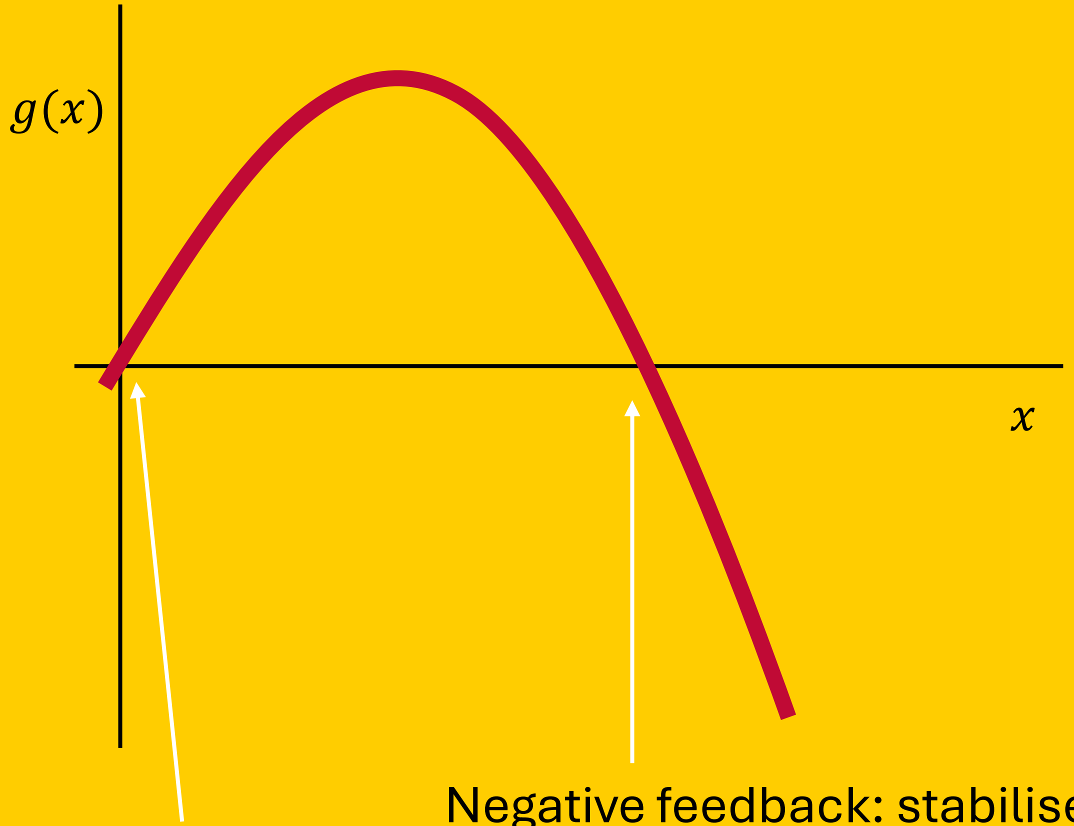
Nonlinear feedbacks



Example:
$f(x) = 1 - x$



$$\frac{dx}{dt}(t) = f(x)x =: g(x)$$



Negative feedback: stabilises

Positive feedback: destabilises

Part III
Mean Field Approximations

Connecting micro to macro

MODEL

state $s_j(t)$

0 : no vegetation

1 : vegetation

Windblown

$$\mathbb{P}(0 \to 1) = \epsilon$$

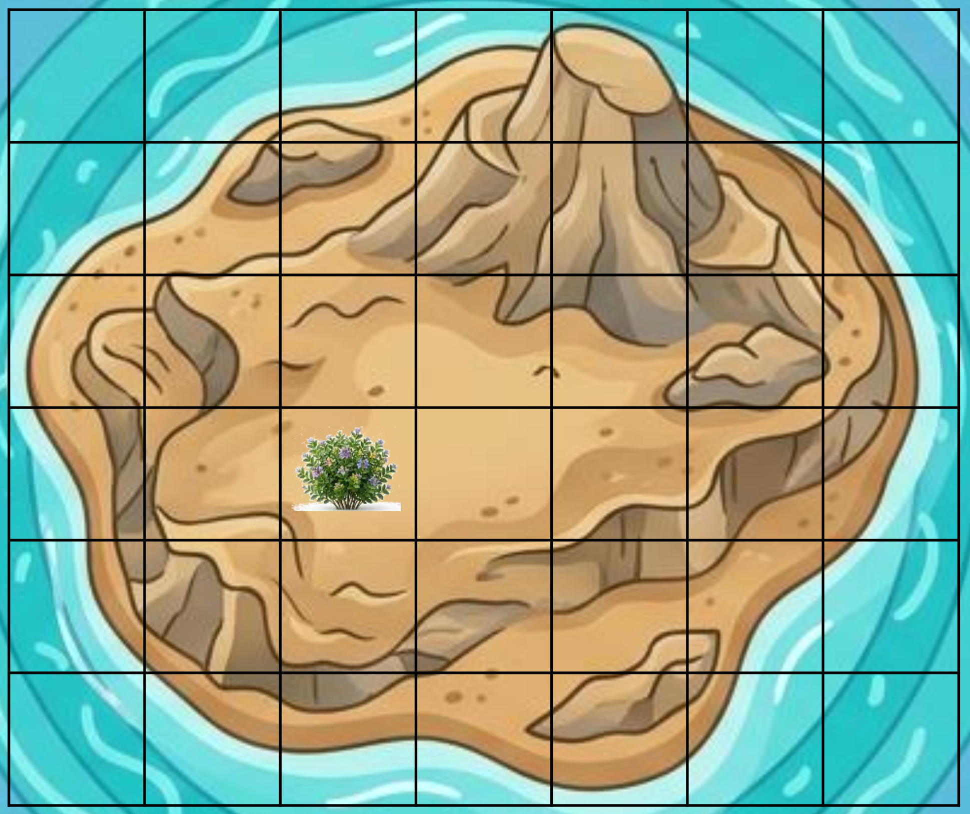
Reproduction

$$\mathbb{P}(0 \to 1) = \alpha\, \rho^2$$

with ρ density around site

Mortality

$$\mathbb{P}(1 \to 0) = \delta$$



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Mean-field approximation

MODEL

state $s_j(t)$

0 : no vegetation

1 : vegetation

Windblown

$$\mathbb{P}(0 \to 1) = \epsilon$$

Reproduction

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Mortality

$$\mathbb{P}(1 \to 0) = \delta$$

Mean field approximation

Let $S(t) := \frac{1}{N}\sum_{j=1}^{N} s_j(t)$ and let $x(t) := \mathbb{E}[S(t)]$

Then:

$$\begin{aligned} x(t+1) &= \mathbb{E}[S(t+1)] \\ &= \mathbb{E}[s(t)=1]\cdot\mathbb{P}(1\to 1) + \mathbb{E}[s(t)=0]\cdot\mathbb{P}(0\to 1) \\ &= x(t)\cdot(1-\delta) + \big(1-x(t)\big)\cdot(\epsilon + \alpha\mathbb{E}[\rho^2]) \\ &= x(t)\cdot(1-\delta) + \big(1-x(t)\big)\cdot(\epsilon + \alpha\, x^2) \end{aligned}$$

In which we assume spatial mixing, i.e. $\mathbb{E}[\rho^2] = x^2$

So: $\boldsymbol{\frac{dx}{dt} = \epsilon - (\epsilon + \delta)x + \alpha x^2 - \alpha x^3}$

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Analyis of mean-field equation

$$\frac{dx}{dt} = \epsilon - (\epsilon + \delta)x + \alpha x^2 - \alpha x^3$$

Using dimensional analysis this reduces to

$$\frac{dy}{d\tau} = \nu \ - y + y^2 - r\, y^3$$

with

$$\nu := \frac{\epsilon\alpha}{(\epsilon+\delta)^2}, r := \frac{\epsilon+\delta}{\alpha}, \mathrm{y} := \frac{\mathrm{x}}{\mathrm{r}}, \tau := \frac{t}{\epsilon+\delta}$$

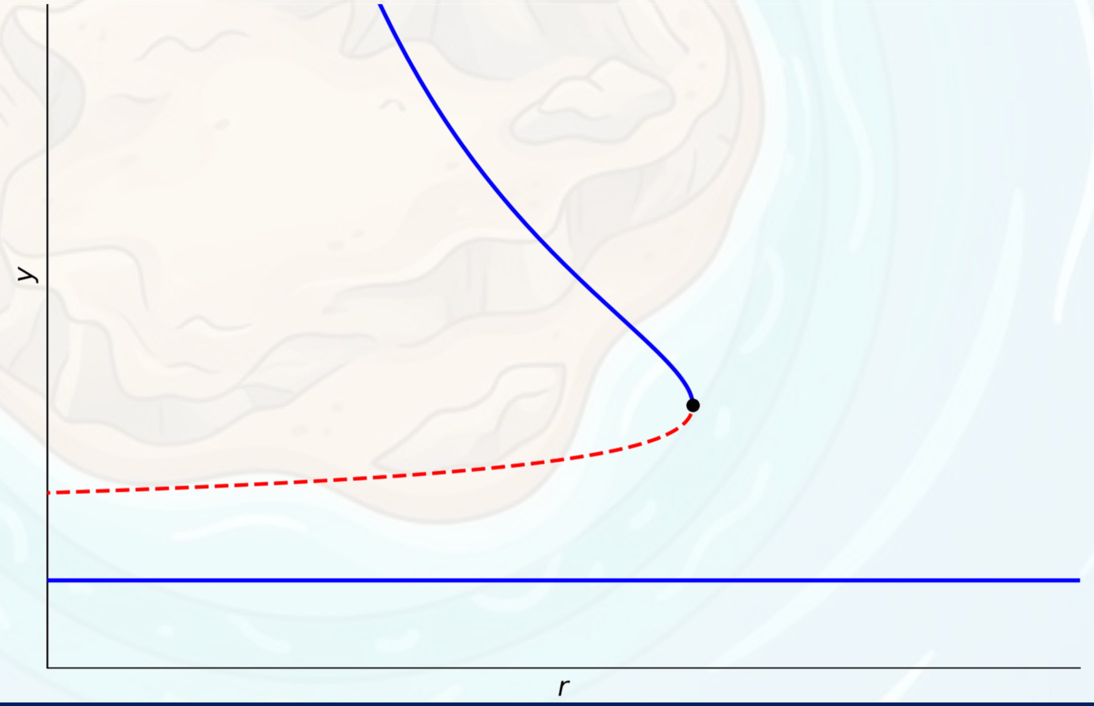
We take $\nu = 0$ ($\epsilon = \ 0$) and study

$$\frac{dy}{d\tau} = -y + y^2 - r\, y^3$$

$$\frac{dy}{d\tau} = -y + y^2 - r\, y^3$$

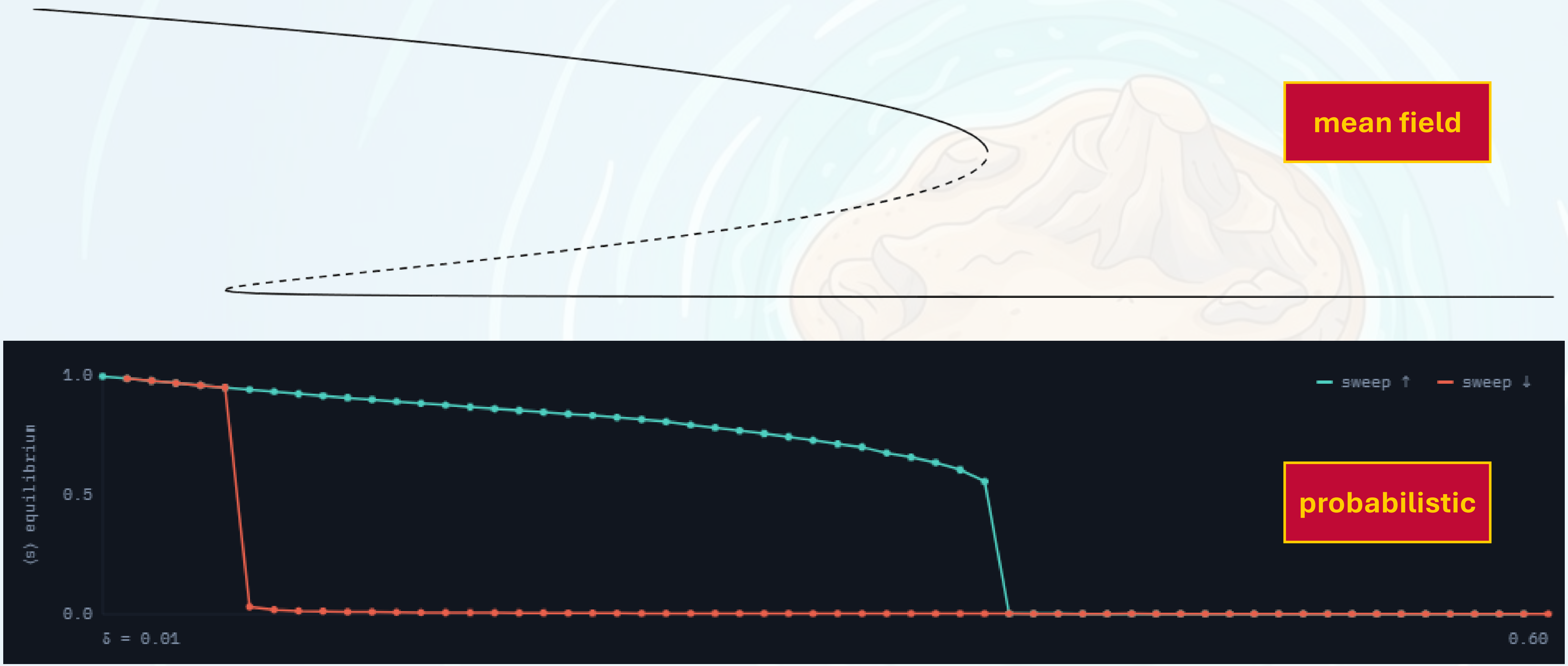
Fixed points:

$$y_1 = 0,\ y_{2,3} = \frac{1 \pm \sqrt{1-4r}}{2r}$$

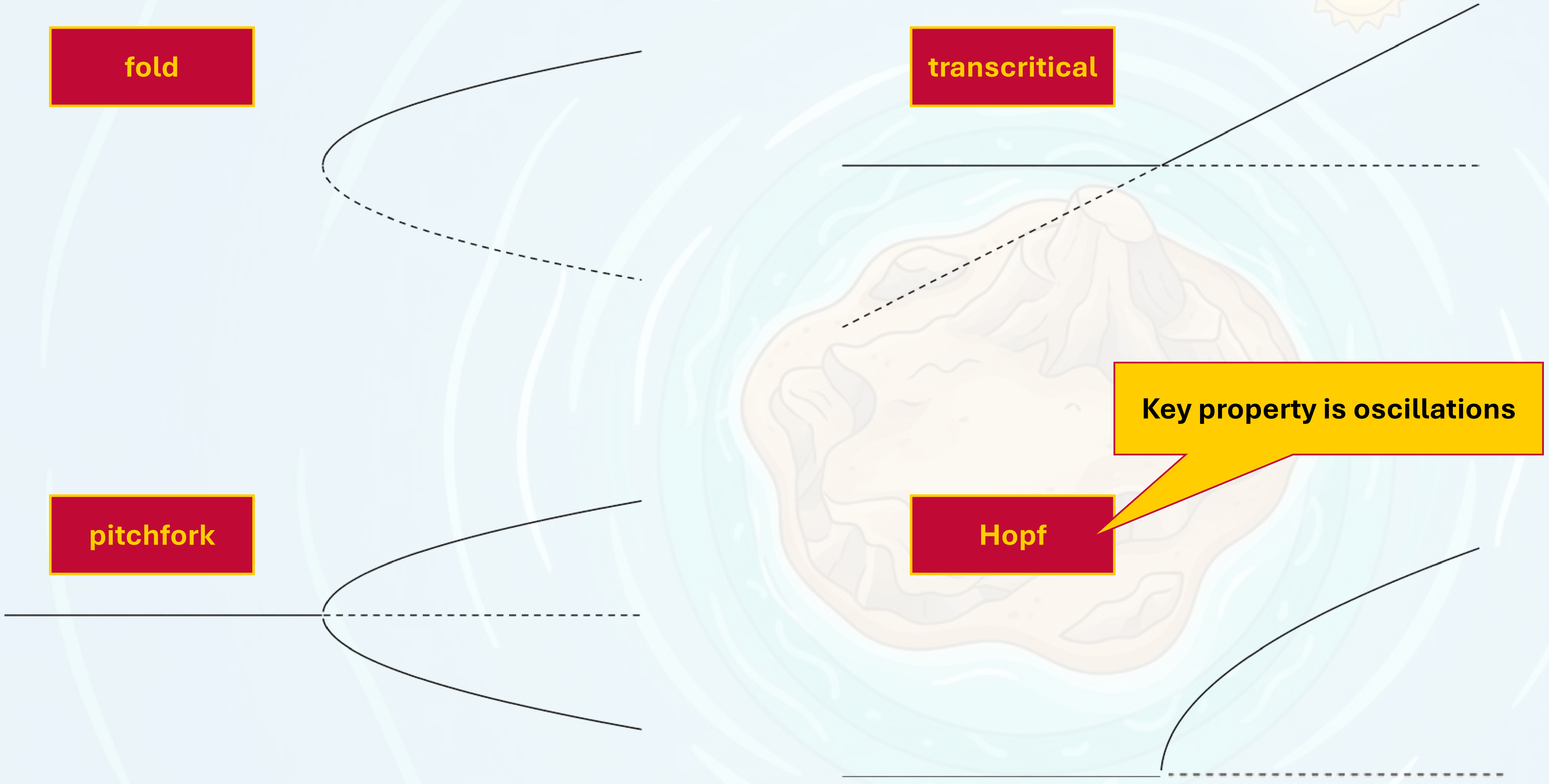


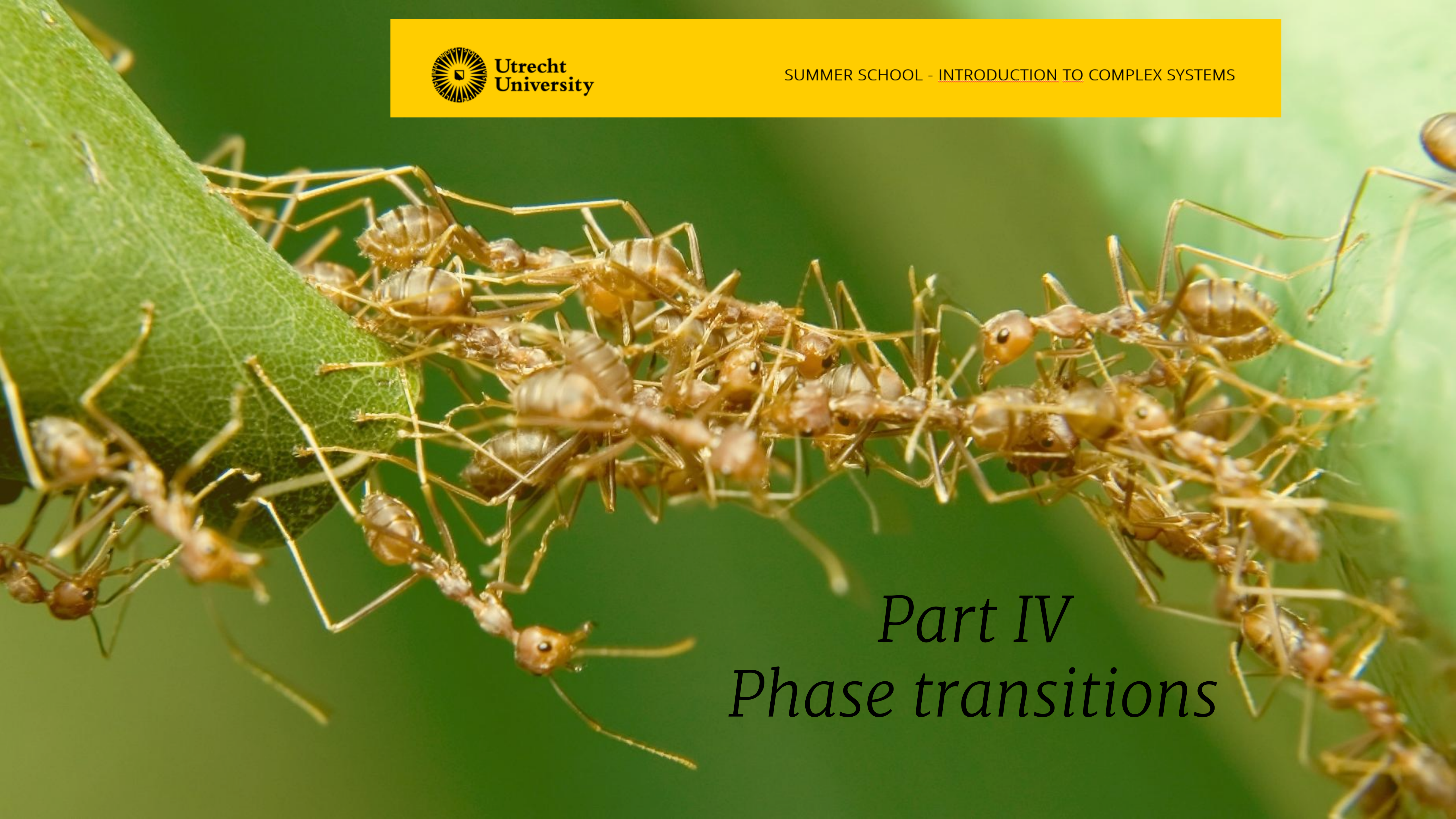
Compare to simulation

Parameters:
$\epsilon = 0.001, \alpha \coloneqq 1.50$



Other bifurcations





Part IV
Phase transitions

Simplified Ising Model

MODEL

state $s_j(t)$
1 : spin up
-1 : spin down

Update rule

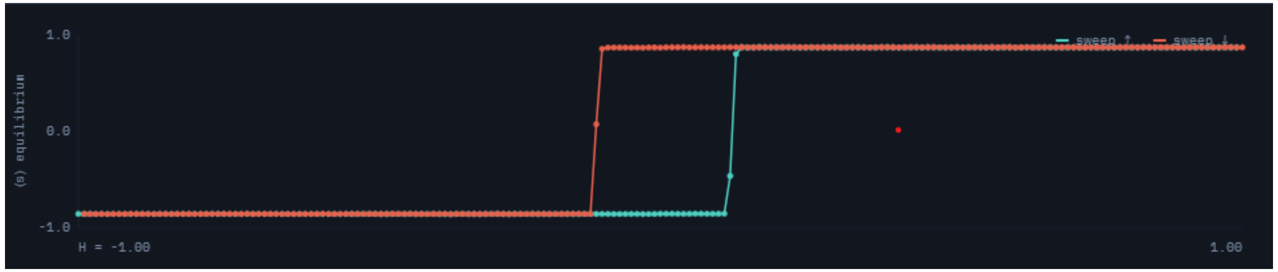
- with probability P: flip cell
- Otherwise:
  - count spin of neighbours
  - add bias H
  - If sum is ≥ 0: cell will be +1
  - If sum is < 0: cell will be -1

Varying P
$(H = 0)$

second-order phase transition

Varying H
$(P = 0.14)$

first-order phase transition



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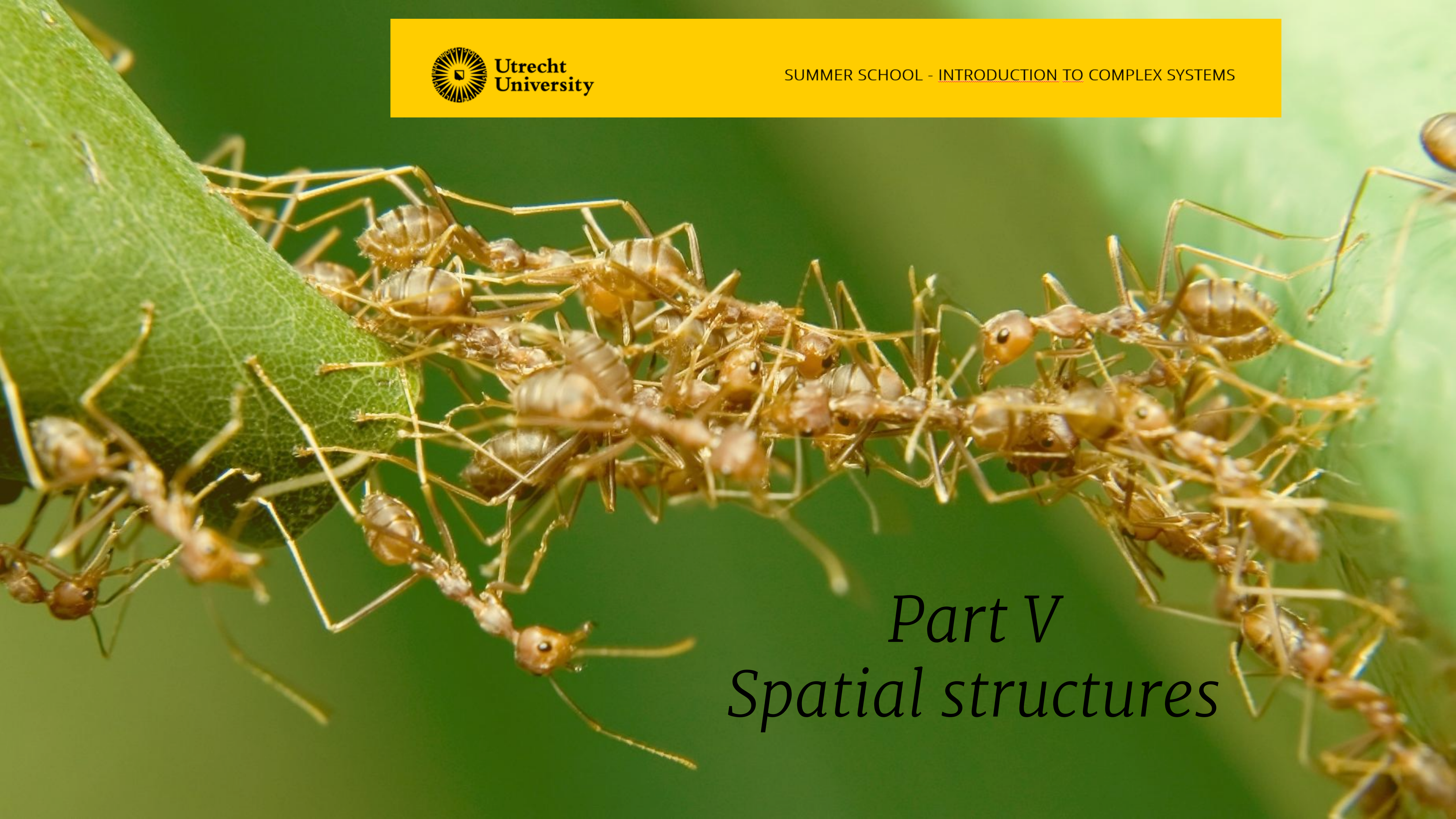


Part V
Spatial structures

Conway's Game of Life

MODEL

state $s_j(t)$

$1:$ ON

$0:$ OFF

Update rule

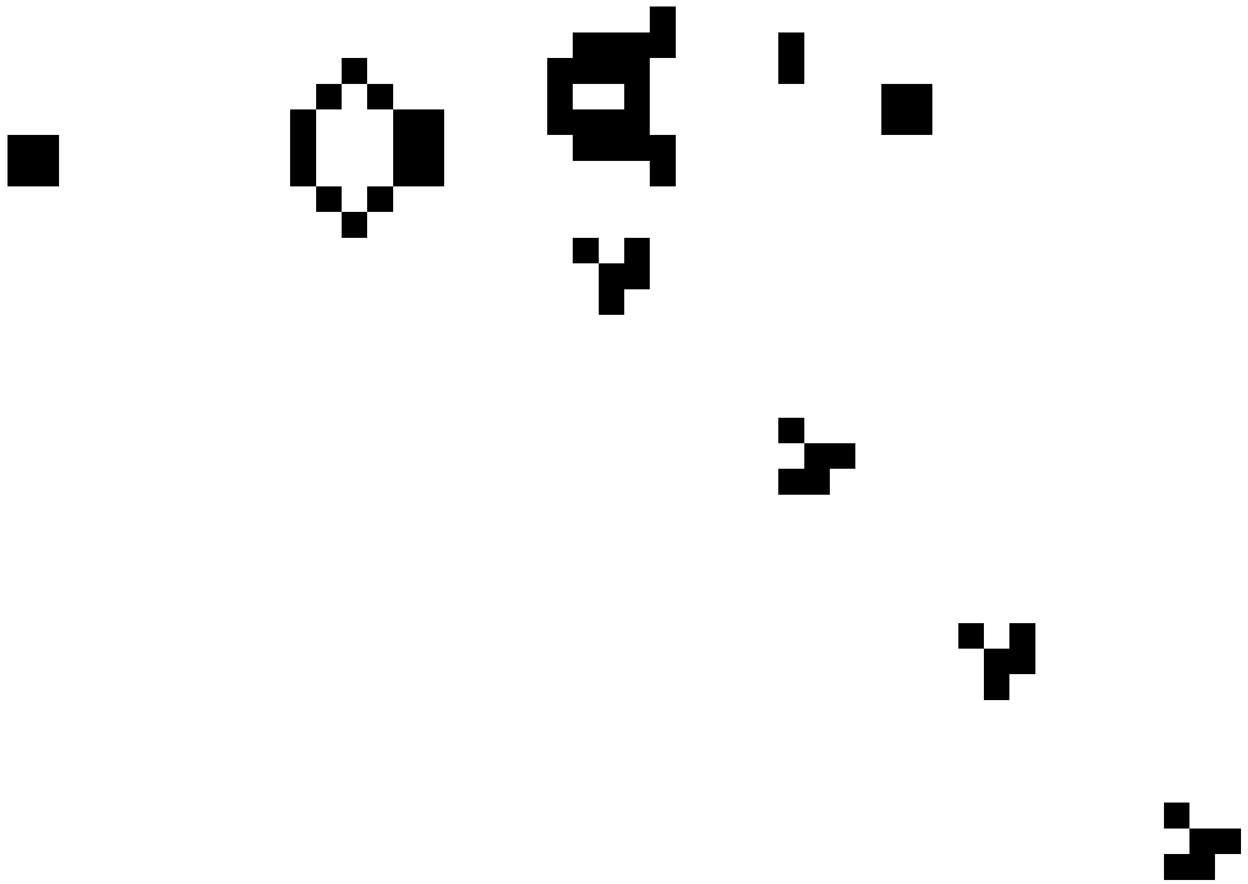
ON → OFF:

- if <2 ON neighbours
- if >3 ON neighbours

OFF → ON:

- if 3 ON neighbours

(otherwise cell does not change)



Implemented at bastiaansen.github.io/CCSS_CA.html

Turing patterning

MODEL

state $s_j(t)$

1 : ALIVE

0 : DEAD

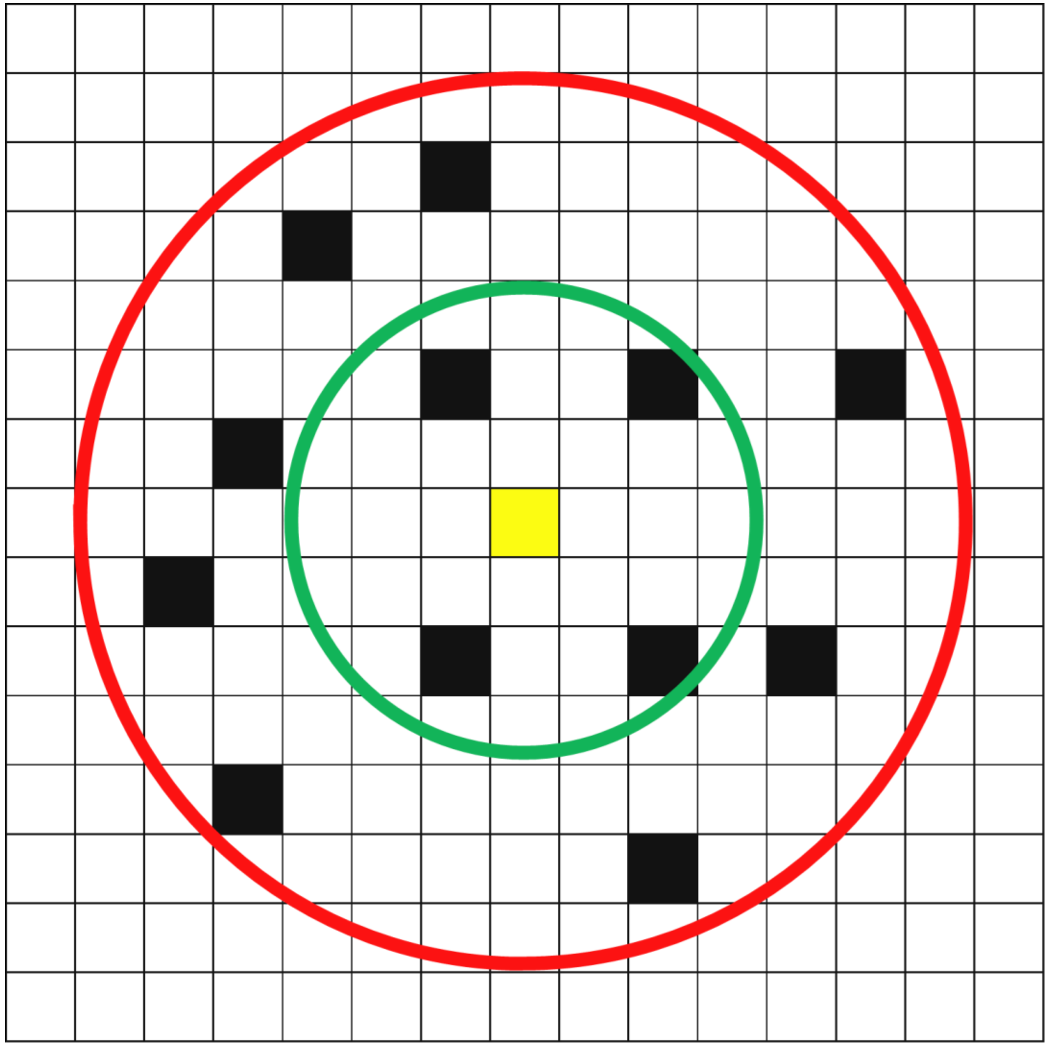
Update rule

Per cell, compute

$$S := \sum_{\mathcal{N}_a} w_a \frac{s_j}{|\mathcal{N}_a|} - \sum_{\mathcal{N}_i} w_i \frac{s_j}{|\mathcal{N}_i|}$$

If $S > 0$: cell becomes ALIVE

If $S \leq 0$: cell becomes DEAD



scale-dependent feedback

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Turing patterning

MODEL

state $s_j(t)$

1 : ALIVE

0 : DEAD

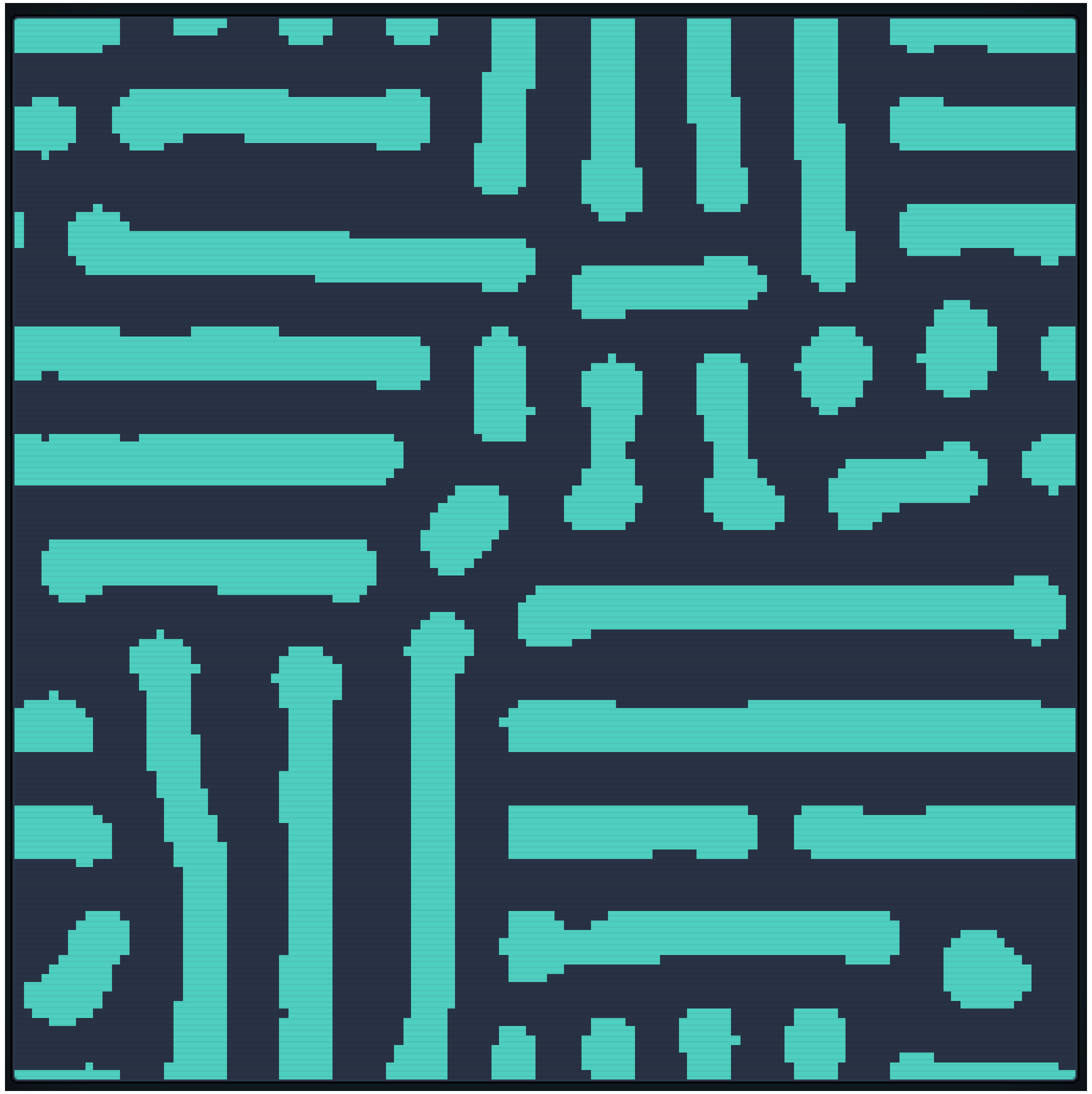
Update rule

Per cell, compute

$$S := \sum_{\mathcal{N}_a} w_a \frac{s_j}{|\mathcal{N}_a|} - \sum_{\mathcal{N}_i} w_i \frac{s_j}{|\mathcal{N}_i|}$$

If $S > 0$: cell becomes ALIVE

If $S \leq 0$: cell becomes DEAD



Parameters: $r_a = 3, w_a = 1{,}15, r_i = 6, w_i = 1.31$

Implemented at bastiaansen.github.io/CCSS_CA.html

Island model (only local interaction)

MODEL

state $s_j(t)$

0 : no vegetation

1 : vegetation

Windblown

$$\mathbb{P}(0 \to 1) = \epsilon$$

Reproduction

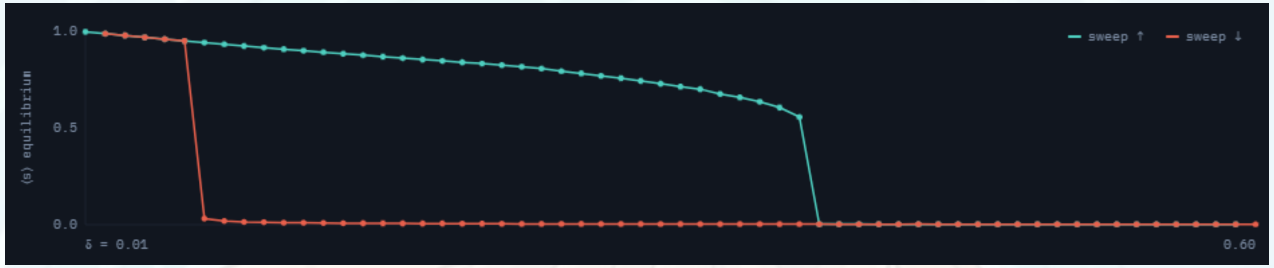
$$\mathbb{P}(0 \to 1) = \alpha\, \rho^2$$

with ρ density around site

Mortality

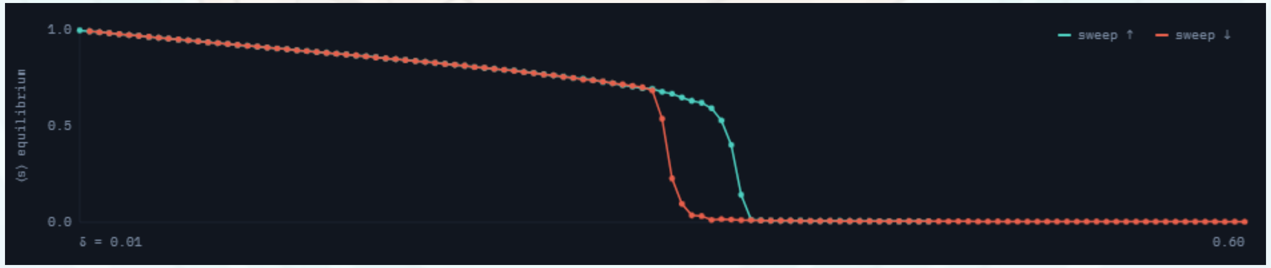
$$\mathbb{P}(1 \to 0) = \delta$$

Global interaction



Parameters: $\epsilon = 0.001, \alpha = 1.50, p = 1$ (use $\delta = 0.33$ to see differences)

Only local interaction



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Part VI
Other modelling frameworks

Agent-based / indivual-based model (ABM/IBM)

MODEL (from agents.jl)

Each agent has location and velocity

Cohesion

- Move towards others

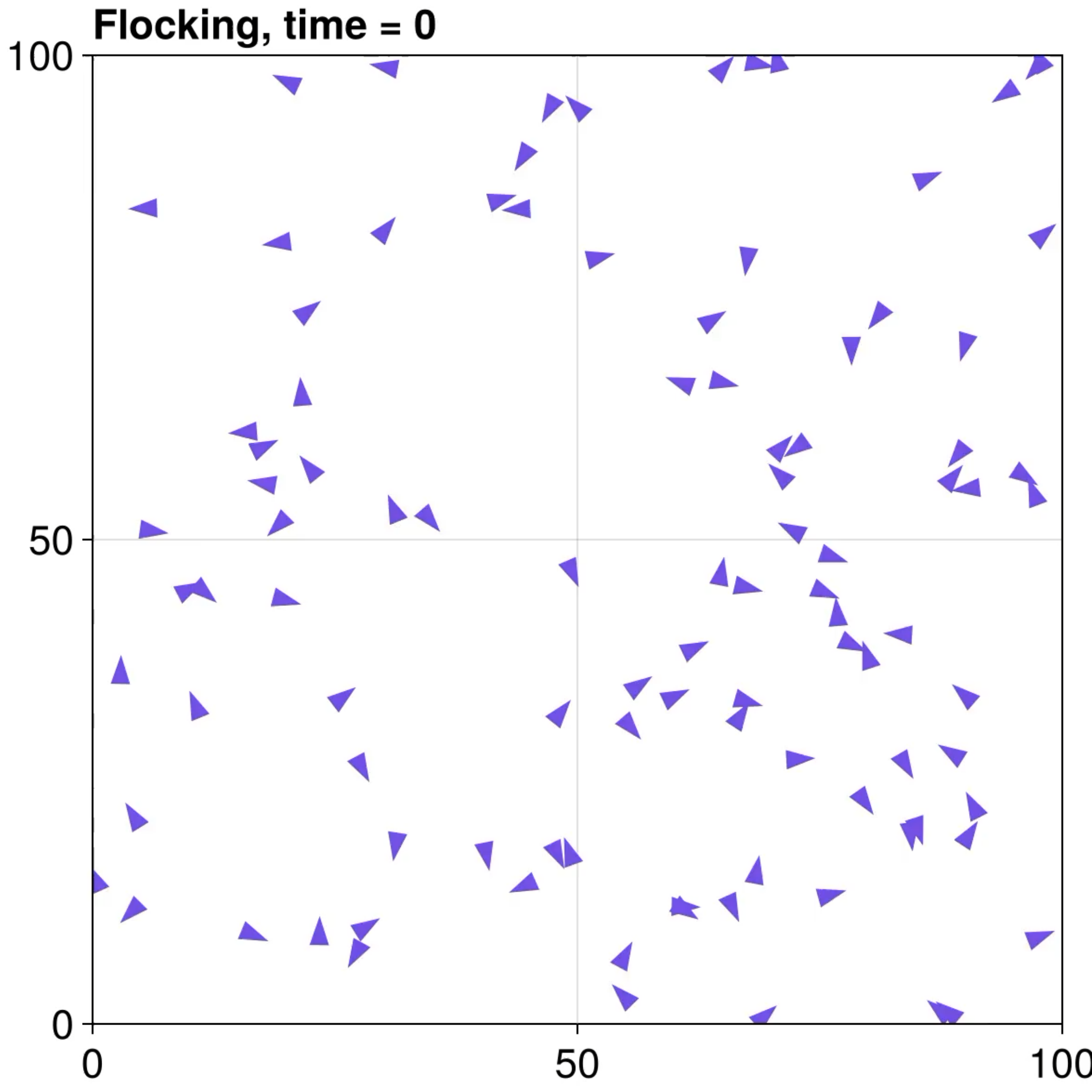
Separation

- Keep minimal distance to others

Alignment

- Match velocity of others

See https://juliadynamics.github.io/Agents.jl/stable/examples/flock/



Partial differential equations

MODEL

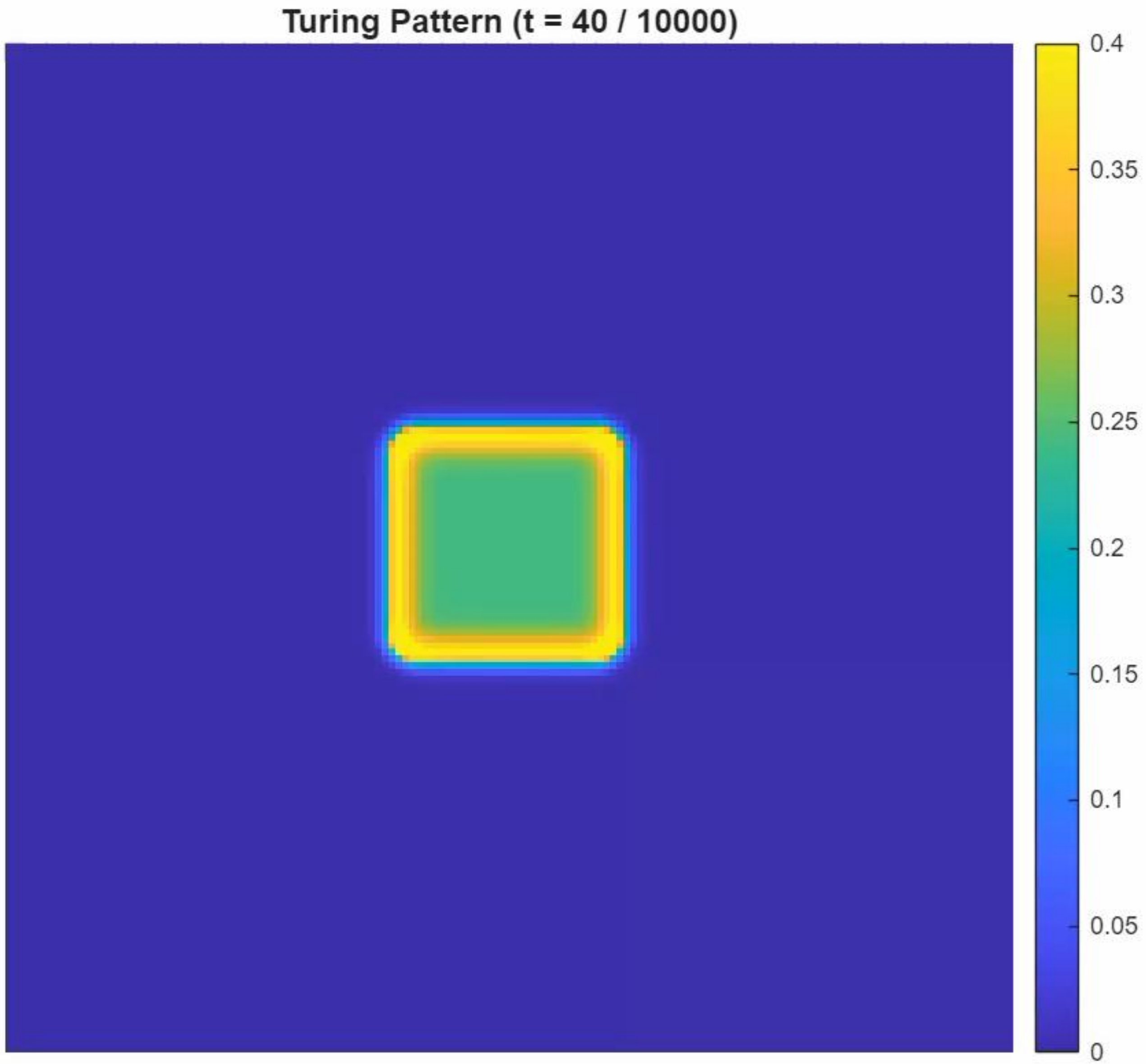
Gray-Scott model

$$\frac{\partial U}{\partial t} = D_u \nabla^2 U - UV^2 + F(1-U)$$

$$\frac{\partial V}{\partial t} = D_v \nabla^2 V + UV^2 - (F+k)V$$

$$D_u = 0.16, D_v = 0.08,$$
$$F = 0.04, k = 0.06$$

Can be derived as spatially-explicit mean-field approximation



Simulation by MATLAB code by Gemini Flash 3.6

Part VII
Ok – so what?

Micro and macro

- Which micro rules lead to observed macro behaviour?
  Examples:
  - How does flocking work?
  - Why are there spatial patterns?

- What macro behaviour emerges from observed micro dynamics?
  Examples:
  - What effects do pheromone deposits have in an ant colony?
  - How does algorithmic trading influence financial markets?

How to study

- Numerical simulations
  - Cellular Automata (CA)
  - Agent-Based Models (ABMs / IBMs)
  - Ordinary or Partial Differential Equations (ODEs & PDEs)

- Analytic results only sometimes available

- Often: system-specific (data) analysis is required

Feedbacks & Emergence

Summary

Feedbacks

negative/stabilising & positive/destabilising

Mean-field approximations

bifurcations/transitions of macro dynamics

Spatial structures

deviations from mean-field approximation

emergence of spatial structures

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